

Fractional SPDE with Time-independent Noise: Solvability and Exact Asymptotics*

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Abstract

In this article, we study the

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1 Introduction

In this paper we study the following fractional *stochastic partial differential equation* (SPDE):

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

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where $a \in (0, 2)$, $b \in (0, 2)$, $r \geq 0$, $\nu > 0$ and $\theta > 0$. Here the noise $W = \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ is a centered and time-independent Gaussian process, defined on a complete probability space $(\Omega, \mathcal{F}, P)$, with mean zero and covariance

$$\mathbb{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi)\overline{\mathcal{F}\psi(\xi)}\mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}},$$

where μ refers to the *spectral measure*, which is assumed to be a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$. Let γ be the Fourier transform of μ , which is also a nonnegative and nonnegative measure on $\mathbb{R}^d$ thanks to Bochner's theorem. Throughout of the paper, we use $\mathcal{F}\varphi(\xi) = \int_{\mathbb{R}^d} \exp(-ix\xi)\varphi(x)dx$ to denote the Fourier transform of a test function φ .

In (1.1), $(-\Delta)^{a/2}$ refers to the *fractional Laplacian* of order a , ∂_t^b denotes the *Caputo fractional differential operator*

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^m(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

where m is an integer, and I_t^r refers to the *Riemann-Liouville fractional integral* of order $r > 0$

$$I_t^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s)ds, \quad \text{for } t > 0,$$

with the convention that when $r = 0$, $I_t^0 = \text{Id}$ reduces to the identity operator. The fundamental solutions to (1.1) consist of a triplet:

$$\left\{ Z_{a,b,r,\nu,d}(t, x), Z_{a,b,r,\nu,d}^*(t, x), Y_{a,b,r,\nu,d}(t, x) : t > 0, x \in \mathbb{R}^d \right\}.$$

Each member of the triplet can be expressed explicitly in terms of the *Fox H-function* which is much more complicated than the fundamental function for either the heat or wave equation; See Theorem 4.1 of [5]. Since we are interested in the constant one initial condition (and zero initial velocity when $b > 1$), Theorem 4.1 (*ibid.*) implies that the corresponding solution to the homogeneous equation (i.e. the solution when there is no driving source) is equal to the constant one. Hence, by setting $G(t, x) := Y_{a,b,r,\nu,d}(t, x)$, (1.1) can be written as the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds, \quad (1.2)$$

E:spdeint

where the stochastic integral is in the *Skorohod* sense; see Definition 2.1 below. In the following, the fundamental solution will exclusively refer to $G(t, x)$, which is indeed a smooth function for $x \neq 0$. Our results rely on the following assumption for the nonnegativity of $G(t, x)$, which is a nontrivial fact; see Remark 1.9 for sufficient conditions.

Nonnegative

Assumption 1.1 (Nonnegativity). Assume that the fundamental solution $G(t, x)$ is nonnegative for all $t > 0$ and $x \in \mathbb{R}^d$.

Regarding the noise, we formulate the following assumption in order to cover the Riesz kernel case, the fractional noise and a mixture of them:

A:Noise

Assumption 1.2 (Noise). Let $k \in \{1, \dots, d\}$ and partition the d -coordinates of $x = (x_1, \dots, x_d)$ into k distinct groups of size d_i so that $d_1 + \dots + d_k = d$. Denote $x_{(i)} = (x_{i_1}, \dots, x_{i_{d_i}})$ to be the coordinates in the i^{th} partition. Assume that the correlation function of the Gaussian noise is given by

$$\gamma(x) = \prod_{i=1}^k |x_{(i)}|^{-\alpha_i} \quad \text{with } \alpha_i \in (0, d_i). \quad (1.3) \quad \text{E:corfuncti}$$

Define $\alpha := \sum_{i=1}^k \alpha_i$.

In a recent work by Balan *et al* [1], the same equation as (1.1), but exclusively for the *stochastic wave equation* (SWE), namely, the case when $a = b = \nu = 2$ and $r = 0$, has been studied, where both the well-posedness and the exact moment asymptotics have been obtained. The corresponding *stochastic heat equation* (SHE), namely, the case when $a = 2$, $b = \nu = 1$ and $r = 0$, has been earlier studied by Hu [9], but only for the well-posedness and exclusively for the fractional noise. The corresponding moment asymptotics have been conjectured in Remark 1.9 of Balan *et al* [1] by *formally* setting $H_0 = 1$ in X. Chen [6]. Note that in X. Chen (*ibid.*), H_0 is strictly in between the open interval $(1/2, 1)$. The case $H_0 = 1$ requires a separate treatment. In this paper, by working on a more general class of SPDEs, we confirm this conjecture on the asymptotics for SHE. Moreover, our results interpolate the asymptotics for both SWE and SHE; see Remark 1.8 below.

The moment asymptotics obtained by X. Chen, such as those in [6, 7], rely crucially on the Feynman-Kac representation of the moments of the solution. However, whenever $b \neq 1$, especially for the case when $b \in (1, 2)$, we are not aware of any such Feynman-Kac formula for the moments. Instead, in the recent work by Balan *et al* [1], this difficulty has been overcome by studying the Wiener chaos expansion of the solution. In this paper, we follow the same strategy laid out by Balan *et al* (*ibid.*).

We need to introduce the following variational constants (see Section 2.2 for more details):

$$\mathcal{M}_a(f, \theta) := \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, f \rangle_{L^2(\mathbb{R}^d)}^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \quad \text{and} \quad \mathcal{M}_a(f) := \mathcal{M}_a(f, 1).$$

For γ defined in (1.3), we use the convention that $\mathcal{M}_a := \mathcal{M}_a(\gamma, 1)$.

Now let us state our main results of the paper. The first main result deals with the well-posedness of the SPDE (1.1) (or (1.2)) as stated in the following theorem:

T:solvable

Theorem 1.3 (Solvability). Assume that both Assumptions 1.1 and 1.2 hold.

(1) (1.1) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$, $t > 0$, and $x \in \mathbb{R}^d$ provided that

$$0 < \alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right). \quad (1.4) \quad \text{E:alpha}$$

(2) Otherwise, if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b} [2(b+r) - 1] \leq d, \quad (1.5) \quad \text{E:critical}$$

then we have the following two cases:

(2-i) For any $p \geq 2$, (1.1) has a unique solution $u(t, x)$ in $L^p(\Omega)$ for all $p \geq 2$ and $x \in \mathbb{R}^d$, but only for $t \in (0, T_p)$ where

$$T_p := \frac{1}{2\theta(p-1)\nu^{-\alpha/a} \mathcal{M}_a^{(2a-\alpha)/a}}. \quad (1.6) \quad \text{E:Tp}$$

(2-ii) For any $t > T_2$, the series (2.11) below diverges, that is, the $L^2(\Omega)$ -solution $u(t, x)$ to (1.1) does not exit whenever $t > T_2$.

The second main result of the paper is about the moment asymptotics. We use $\|\cdot\|_p$ to denote the $L^p(\Omega)$ moments.

T:asym **Theorem 1.4.** Under Assumptions 1.1 and 1.2, if condition (1.4) holds, then we have that

$$\lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p = \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right), \quad (1.7) \quad \text{E:momentasy}$$

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p := (p-1)^{1-1/\beta} t. \quad (1.8) \quad \text{E:beta\&tp}$$

Proof. We prove the matching upper bound (4.1) and the lower bound (5.1) of (1.7) at the end of Sections 4 and 5 below, respectively, which together prove (1.7). $\square$

As a direct consequence of (1.7), one can send either t or p to infinity as follows:

C:Lim **Corollary 1.5.** Under both Assumptions 1.1 and 1.2, if condition (1.4) holds, then

(1) For all $p \geq 2$ fixed, it holds that

$$\lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) = p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right); \quad (1.9) \quad \text{E:momentasy}$$

(2) For all $t > 0$ fixed, it holds that

$$\lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E}(|u(t, x)|^p) = t^\beta \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \quad (1.10) \quad \text{E:momentasy}$$

Before the end of this Introduction, let us make a few remarks:

R:Noise **Remark 1.6.** Note that the spectral density corresponding to (1.3) is given by

$$\mu(d\xi) = \varphi(\xi)d\xi \quad \text{with} \quad \varphi(\xi) = \prod_{i=1}^k C_{\alpha_i, d_i} |\xi_{(i)}|^{-(d_i - \alpha_i)}. \quad (1.11) \quad \text{E:specfunct}$$

Moreover, one can find a nonnegative and nonnegative definite K such that $\gamma = K * K$ where ‘ $*$ ’ denotes the spatial convolution. Indeed, one can choose

$$K(x) = \prod_{i=1}^k \beta_{\alpha_i, d_i} |x_{(i)}|^{-(d_i + \alpha_i)/2}. \quad (1.12) \quad \text{E:K}$$

The two constants in both (1.11) and (1.12) are defined as

$$C_{\alpha,d} = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)} \quad \text{and} \quad \beta_{\alpha,d} = \pi^{-d/4} \frac{\Gamma((d+\alpha)/4)}{\Gamma((d-\alpha)/4)} \sqrt{\frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)}}. \quad (1.13)$$

E:noisecons

Remark 1.7. We have the following special cases: (1) The case $k = 1$ in (1.3) (or (1.11)) recovers the Riesz kernel case. In this case,

$$\gamma(x) = |x|^{-\alpha}, \quad \varphi(x) = C_{\alpha,d} |x|^{-(d-\alpha)} \quad \text{and} \quad K(x) = \beta_{\alpha,d} |x|^{-(d+\alpha)/2}.$$

(2) The case when $k = d$ in (1.3) (or (1.11)) recovers the time-independent fractional noise. The corresponding SHE with such noise was earlier studied by Hu [9]. For this noise, we have that

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}, \quad \varphi(\xi) = \prod_{i=1}^d C_{\alpha_i,1} |\xi_i|^{-(1-\alpha_i)} \quad \text{and} \quad K(x) = \prod_{i=1}^d \beta_{\alpha_i,1} |x_i|^{-(1+\alpha_i)/2}.$$

Remark 1.8. The asymptotics in (1.7) interpolates the corresponding results for the stochastic wave and heat equations:

(1) By setting

$$a = b = \nu = 2 \quad \text{and} \quad r = 0,$$

we have that

$$\beta = \frac{4-\alpha}{3-\alpha} \quad \text{and} \quad t_p = (p-1)^{1/(4-\alpha)} t,$$

and results in (1.7), (1.9), and (1.10) recover the corresponding results for the stochastic wave equation, namely, (1.9), (1.10), and (1.11) of [1], respectively.

(2) As for the stochastic heat equation case, we are not aware of the exact asymptotics results for SHE with time-independent noise case. However, as explained in Remark 1.9 of [1], one may apply Chen's asymptotic results in [6] for SHE with time-dependent noise, by formally setting $H_0 = 1$, to obtain the corresponding time-independent results. In particular, by setting

$$a = b = 2, \quad \nu = 1 \quad \text{and} \quad r = 0,$$

we have that

$$\beta = \frac{4-\alpha}{2-\alpha} \quad \text{and} \quad t_p = (p-1)^{2/(4-\alpha)} t,$$

and results in (1.7) and (1.9) recover the corresponding results for the stochastic heat equation, namely, (1.16) and (1.17) of [1], respectively.

Remark 1.9. Regarding the nonnegativity assumption — Assumption 1.1, thanks to Theorem 4.6 of [5], we have the following four groups of sufficient conditions, under either group of which $G(t, \cdot)$ is nonnegative (see Fig. 1 for an illustration):

1. $d \geq 1, b \in (0, 1), a \in (0, 2], r \geq 0$;
2. $d \geq 1, b = 1, a \in (0, 2], r = 0$ or $r > 1$;
3. $1 \leq d \leq 3, 1 < b < a \leq 2, r > 0$;
4. $1 \leq d \leq 3, 1 < b = a < 2, r > \frac{d+3}{2} - b$.

In the following, we first give some preliminaries in Section 2 including the Malliavin calculus, the definition of the solution, and some variational constants. The existence and uniqueness of a $L^p(\Omega)$ solution — Theorem 1.3 — is proven in Section 3. The upper bound and lower bound for (1.7) are established in Sections 4 and 5, respectively. Finally, in the appendix — Section 6, we list a few proofs of the results in the paper.

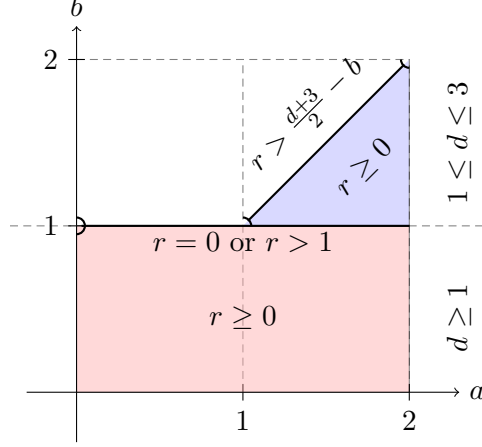


Figure 1: Illustration of the sufficient conditions (Remark 1.9) for $G(t, \cdot)$ to be nonnegative.

2 Some preliminaries

2.1 Malliavin Calculus

To briefly summarize [8], we start with a nonnegative and nonnegative definite tempered measure Γ with density γ in the sense that $\Gamma(dx) = \gamma(x)dx$, and

$$\int_{\mathbb{R}^d} \Gamma(dx) (\varphi * \tilde{\varphi})(x) \geq 0 \quad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^d)$$

where $\tilde{\varphi}(x) = \varphi(-x)$. In addition we require the existence of an $r > 0$ such that

$$\int_{\mathbb{R}^d} \Gamma(dx) \frac{1}{(1 + |x|^2)^r} < \infty.$$

According to the Bochner theorem, there exists a nonnegative measure μ , often referred to as the spectral measure on $\mathbb{R}^d$ whose Fourier transform is Γ in the sense that for any $\varphi \in \mathcal{S}(\mathbb{R}^d)$

$$\int_{\mathbb{R}^d} \Gamma(dx) \varphi(x) = \int_{\mathbb{R}^d} \mu(dt) \mathcal{F}\varphi(t)$$

The functional defined on the space of test functions $\mathcal{D}(\mathbb{R}^d)$

$$C(\varphi, \psi) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi)$$

is nonnegative definite and thus we may associate it as the covariance functional of a zero-mean Gaussian processes, $W := \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$. In other words,

$$\mathbb{E}(W(\varphi)W(\psi)) = \int_{\mathbb{R}} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}}.$$

Let $\mathcal{H}$ be the completion of $\mathcal{D}(\mathbb{R}^d)$ with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and thus we see $\varphi \mapsto W(\varphi)$ is an isometry from the normed space $(\mathcal{D}(\mathbb{R}^d), \|\cdot\|_{\mathcal{H}})$ to $L^2(\Omega)$ where $\|f\|_{\mathcal{H}} = \langle f, f \rangle_{\mathcal{H}}^{1/2}$.

We denote δ the Skorohod integral with respect to W and we denote its domain $\text{Dom}(\delta)$. For any $u \in \text{Dom}(\delta)$, we write

$$\delta(u) = \int_{\mathbb{R}^d} u(x) W(dx).$$

Through construction of δ , $\delta(u)$ is a random variable in $L^2(\Omega)$. For a deeper explanation, we direct the reader to Nualart [11].

D:Sol

Definition 2.1. A random field $u = \{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$ such that $\|u(t, x)\|_2 < \infty$ for all $t \geq 0$ and $x \in \mathbb{R}^d$ is called a *mild solution* to the equation (1.1) if the random field $\{G(t-s, x-y)u(s, y) : s \in [0, t], y \in \mathbb{R}^d\}$ is Skorohod integrable for each $s \in [0, t]$ and $x \in \mathbb{R}^d$ fixed and satisfies the following integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y)u(s, y)W(\delta y) \right) ds. \quad (2.1)$$

Through the standard Picard iteration scheme, the solution can be expressed by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.2)$$

E:solwexp

where $I_k : \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k) \rightarrow \mathcal{H}_k$ is the k^{th} order Skorohod integral and $\mathcal{H}_k$ is the k^{th} Wiener chaos space. The kernels $f_k(\cdot, x, t) \in \mathcal{H}^{\otimes k}((\mathbb{R}^d)^k)$ are calculated through the Piccard iteration scheme to be

$$\begin{aligned} f_n(x_1, \dots, x_n; x, t) &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) dt_1 \cdots dt_n \\ &= \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) dt_1 \cdots dt_n. \end{aligned} \quad (2.3)$$

E:wker

For ease of notation throughout this article, we write the above integrals as

$$\begin{aligned} f_n(x_1, \dots, x_n; x, t) &= \int_{[0, t]_{<}^n} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) d\vec{t} \\ &= \int_{[0, t]_{<}^n} G(t_1, x-x_n) \cdots G(t_n-t_{n-1}, x_2-x_1) d\vec{t}. \end{aligned}$$

where in general $d\vec{z} = \prod_{k=1}^d dz_k$ and the dimension will always be easily understood. We denote by $\widetilde{f}_n(\cdot, x, t)$ to be the symmetrization of $f_n(\cdot, x, t)$ in the variables $x_1, \dots, x_n$:

$$\widetilde{f}_n(\cdot, x, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} f_n(x_{\rho(1)}, \dots, x_{\rho(n)}) \quad (2.4)$$

$$= \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} G(t-t_n, x-x_{\rho(n)}) \cdots G(t_2-t_1, x_{\rho(2)}-x_{\rho(1)}) d\vec{t} \quad (2.5)$$

where Σ_n is the set of all permutations of $\{1, \dots, n\}$. In particular, for $x = 0$ we have

$$\widetilde{f}_n(\cdot; 0, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{[0, t]_{<}^n} G(t_1, x_{\rho(1)}) \cdots G(t_n-t_{n-1}, x_{\rho(n)}-x_{\rho(n-1)}) d\vec{t}. \quad (2.6)$$

E:fperm0

By setting $t_{n+1} = t$, the Fourier transform of the kernels, f_n , is given by

$$\mathcal{F}f_n(\cdot; x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0, t]_{<}^n} \overline{\prod_{k=1}^n \mathcal{F}G(t_{t_{k+1}} - t_k, \cdot)} \left(\sum_{j=1}^n \xi_j \right) dt. \quad (2.7)$$

E:fweker

Recall the notation above that $G(t, x) = Y_{a,b,r,d}(t, x)$. The following result shows an important scaling property that both $\mathcal{F}G(t, \cdot)(\xi)$ and $\left\| \widetilde{f}_n(\cdot, t, x) \right\|_{\mathcal{H}^{\otimes n}}$ have.

L:Scale

Lemma 2.2. *For any $c > 0$ and $t > 0$, the following scaling properties hold:*

$$\mathcal{F}G(t, \cdot)(c\xi) = c^{-\frac{a}{b}(b+r-1)} \mathcal{F}G\left(c^{\frac{a}{b}}t, \cdot\right)(\xi) \quad \text{and} \quad \mathcal{F}G(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}G(t, \cdot)(c^{b/a}\xi), \quad (2.8)$$

E:green scale

$$\mathcal{F}\widetilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F}\widetilde{f}_n(\cdot, 0, t)(c^{b/a}\xi_1, \dots, c^{b/a}\xi_n), \quad (2.9)$$

E:Ffnscale

$$\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (2.10)$$

E:normscale

Proof. The scaling properties in (2.8) is a direct consequence of the explicit expression of $\mathcal{F}G(t, \cdot)(\xi)$ as in (4.8) of [5]. Property (2.9) is an easy exercise of change of variables. Property (2.10) is a direct consequence of (2.8), (2.9), and the scaling property of the spectral measure μ . We leave the details for the interested readers. $\square$

Recall the following standard result about the existence and uniqueness of the solution when the solution can be written as the Wiener chaos expansion as in (2.2).

E:2momseries

Theorem 2.3. *Fix any $T \in (0, \infty]$. Suppose that $f_n(\cdot, x, t) \in \mathcal{H}^{\otimes n}$ for any $t \in (0, T)$, $x \in \mathbb{R}^d$ and $n \geq 1$. Then equation (1.1) has a unique $L^2(\Omega)$ solution on $(0, T) \times \mathbb{R}^d$ if and only if the series (2.2) converges in $L^2(\Omega)$ for any $(t, x) \in (0, T) \times \mathbb{R}^d$, which is equivalent to the convergence of the series (2.11). In this case, the solution is given by (2.2) with the second moment given by*

$$\mathbb{E}(u(t, x)^2) = \sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{for all } (t, x) \in (0, T) \times \mathbb{R}^d. \quad (2.11)$$

E:2mom

2.2 Some asymptotics and variational constants

E:Variation

Recall that the correlation function γ satisfies Assumption 1.2 and the corresponding spectral measure is μ ; see Remark 1.6. Define

$$\rho_{\nu, a}(\gamma) = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \quad (2.12)$$

E:rho

and

$$\begin{aligned} \mathcal{M}_\alpha(\gamma, \theta) &:= \sup_{g \in \mathcal{F}_a} \left\{ \left(\iint_{\mathbb{R}^{2d}} g^2(x)g^2(y)\gamma(x+y)dx dy \right)^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \\ &= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, \gamma \rangle_{L^2(\mathbb{R}^d)} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\}, \end{aligned} \quad (2.13)$$

E:Lambda

where

$$\mathcal{E}_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi, \quad \text{and} \quad (2.14)$$

E:Egg

$$\mathcal{F}_a := \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(f, f) < \infty \right\}. \quad (2.15)$$

E:Fa

We use the convention that $\mathcal{M}_a(\gamma) := \mathcal{M}_a(\gamma, 1)$. By slightly modifying the proof of [1, Lemma 2.3], one can show that

$$\mathcal{M}_a(\Theta\gamma, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(\gamma, 1), \quad \text{for all } \theta \text{ and } \Theta > 0. \quad (2.16)$$

E:ScaleM

For the Riesz kernel case (see Remark 1.7), Bass, Chen and Rosen [3] established that when $a \in (0, 2]$ and $\nu = 2$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{2,a}(|\cdot|^{-\alpha})), \quad (2.17) \quad \text{E:oldlim}$$

and ¹

$$\rho_{2,a}(|\cdot|^{-\alpha}) = \mathcal{M}_a^{2-(\alpha/a)}(|\cdot|^{-\alpha}, 2). \quad (2.18) \quad \text{E:oldrhoVar}$$

We first apply some scaling arguments to accommodate the parameter ν in both (2.17) and (2.18), the proof of which can be found in Appendix:

L:rhoVar

Lemma 2.4 (The Riesz kernel case). *If $\gamma(x) = |x|^{-\alpha}$ for some $\alpha \in (0, d)$, then for any $\nu > 0$ and $a \in (0, 2]$,*

$$\rho_{\nu,a}(|\cdot|^{-\alpha}) = \left(\frac{\nu}{2}\right)^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(|\cdot|^{-\alpha}, 2) = \nu^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(|\cdot|^{-\alpha}), \quad (2.19) \quad \text{E:rhoLam}$$

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu,a}(|\cdot|^{-\alpha})). \quad (2.20) \quad \text{E:LogRhoRie}$$

More generally we have the following:

T:Variation

Theorem 2.5. *For the correlation function γ that satisfies Assumption 1.2, both (2.19) and (2.20) hold with $|\cdot|^{-\alpha}$ and μ replaced by γ and μ as in (1.11), respectively. More precisely, it holds that*

$$\rho_{\nu,a}(\gamma) = \nu^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(\gamma) < \infty, \quad (2.21) \quad \text{E:rhoMrep}$$

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu,a}(\gamma)). \quad (2.22) \quad \text{E:LogRho}$$

Sketch of the proof of Theorem 2.5. The proof of this theorem follows essentially the identical proof as Bass *et al* [3], which is exclusively for the Riesz kernel. One simplification is that we only need to handle the case $p = 2$ thanks to the hypercontractivity property. For our slight extension to the noise given in Assumption 1.2, there is no need to repeat their paper. Instead we will only point out the differences and necessary changes. For your convenience, the correspondence of parameters between Bass *et al* [3] and the current paper is listed in the following Table 1.

Table 1: Notation correspondence.

Tb:Notation

	Laplace		Noise			Moment	Variational Const.
Bass <i>et al</i> [3]	β	2	σ	$ \cdot ^{-\sigma}$	$\varphi_{d-\sigma}$	p	Λ_σ
Current paper	a	ν	α	γ	φ	2	$\mathcal{M}_a(\cdot ^{-\alpha}, 2)$

Theorem 2.5 is proved by showing the following claims: for γ given in Assumption 1.2,

¹In Theorem 1.5 or eq. (1.20) of Bass *et al* [3], the factor $(2\pi)^{-d}$ should not be present.

- (i) $\rho_{\nu,a}(\gamma) < \infty$;
- (ii) $\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \geq \log(\rho_{\nu,a}(\gamma))$;
- (iii) $\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \leq \log(\rho_{\nu,a}(\gamma))$;
- (iv) $\mathcal{M}_a(\gamma) < \infty$;
- (v) $\rho_{\nu,a}(\gamma) = \nu^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(\gamma)$.

Part (i) which corresponds to Lemma 1.6 (*ibid.*) is established by Lemma 2.6 below.

Following exactly the same arguments as those in Section 3 (*ibid.*) with $\varphi_{d-\sigma}$ (*ibid.*) replaced by our φ as in (1.11), one can prove part (ii) for $\nu = 2$. Then an application of the scaling property as the proof of Lemma 2.4 shows the general case $\nu > 0$.

The proof of the upper bound, namely part (iii), is more challenging. This part corresponds to Sections 5 and 6 (*ibid.*). By examining these two sections carefully, we need to make some changes in Section 5 (*ibid.*) and the arguments in Section 6 (*ibid.*) follows unchanged. For Section 5 (*ibid.*), we need to use the following decomposition of φ as opposed to (5.4) (*ibid.*):

$$\varphi(\xi) = \prod_{i=1}^k C_{\alpha_i, d_i} |\xi_{(i)}|^{-(d_i - \alpha_i)} = \prod_{i=1}^k C_{\alpha_i, d_i} (\mathcal{P}_i * \mathcal{P}_i)(\xi_{(i)}),$$

with

$$\mathcal{P}_i(\xi_{(i)}) = \beta_{d_i - \alpha_i, d_i} |\xi_{(i)}|^{-(d_i - \alpha_i/2)};$$

see (1.13) for the constants. Or equivalently,

$$\varphi(\xi) = (\mathcal{P} * \mathcal{P})(\xi) \quad \text{with} \quad \mathcal{P}(\xi) := \prod_{i=1}^k \sqrt{C_{\alpha_i, d_i}} \beta_{d_i - \alpha_i, d_i} |\xi_{(i)}|^{-(d_i - \alpha_i/2)}.$$

Now (5.5) (*ibid.*) should be written as

$$\mathcal{P}_{\beta, \epsilon}(\xi) = \hat{h}(\epsilon \xi) \prod_{i=1}^k \frac{\sqrt{C_{\alpha_i, d_i}} \beta_{d_i - \alpha_i, d_i}}{\beta + |\xi_{(i)}|^{-(d_i - \alpha_i/2)}}, \quad \text{for all } \beta, \epsilon \geq 0,$$

where $h(\cdot)$ is defined in (5.2) (*ibid.*). So $\mathcal{P}(\xi) = \mathcal{P}_{0,0}(\xi)$ and

$$(\mathcal{P}_{\beta, \epsilon} * \mathcal{P}_{\beta, \epsilon})(\xi) \leq (\mathcal{P}_{\beta, 0} * \mathcal{P}_{\beta, 0})(\xi) \leq (\mathcal{P}_{0,0} * \mathcal{P}_{0,0})(\xi) = \varphi(\xi);$$

see (5.6) (*ibid.*). With these changes, one can update accordingly the proof of Lemma 5.1 (*ibid.*) without any difficulty. Then the rest part of Section 5 (*ibid.*) follows unchanged. In this way, we establish part (iii) for $\nu = 2$. Finally, a scaling argument as in part (ii) proves part (iii) for all $\nu > 0$.

Parts (iv) and (v) correspond to Section 7 (*ibid.*). In particular, part (iv) corresponds to Lemma 7.1 (*ibid.*). Note that we only need to study the case $p = 2$. By (2.23) with $\varphi(x - y)$ replaced by $\gamma(x - y)$, we see that

$$\int_{(\mathbb{R}^d)^2} g^2(x) g^2(-y) \gamma(x - y) dx dy \leq C \|\tilde{g}^2\|_{L^{2d/(2d-\alpha)}(\mathbb{R}^d)} \|g^2\|_{L^{2d/(2d-\alpha)}(\mathbb{R}^d)} = C \|g\|_{L^{4d/(2d-\alpha)}(\mathbb{R}^d)}^4,$$

where $\tilde{g}^2(x) = g(-x)$. Thus equation (7.1) (*ibid.*) can be applied in our setting . The rest of the proof of Lemma 7.1 (*ibid.*) remains unchanged.

It remains to update the proof of Theorem 1.5 on Section 7 (*ibid.*). For this, one needs only to update the four appearances of $1/(\cdot)^\sigma$ in (7.15), (7.22) and (7.23) (*ibid.*) to $\gamma(\cdot)$. Note that the factor $(2\pi)^{-d(p+1)}$ in the first equation of (7.22) (*ibid.*) should be $(2\pi)^{-dp}$. With this, we complete the sketch proof of Theorem 2.5. $\square$

The proof of Lemma 1.6 (*ibid.*) relies on inequality (1.27) on p. 630 (*ibid.*), which was a consequence of Sobolev's inequality. For the more general noises studied in this paper, we can no longer apply this inequality. Instead, we prove the following lemma using the weak Young's inequality as a generalization of Lemma 1.6 (*ibid.*).

Lemma 2.6. *For any f, g, h with $h \geq 0$, for φ given as in (1.11) (see also Assumption 1.2), and for all $p \geq 2$, it holds that*

$$\left(\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{|f(x+y)g(y)|}{\sqrt{h(x+y)}\sqrt{h(y)}} dy \right]^p \varphi(x) dx \right)^{1/p} \leq C \|f\|_{L^2(\mathbb{R}^d)} \|g\|_{L^2(\mathbb{R}^d)} \|h^{-1}\|_{L^{pd/\alpha}(\mathbb{R}^d)}.$$

Note that we only need the case $p = 2$. Nevertheless, in this lemma we prove a general case for all $p \geq 2$.

Proof. By observing the proof of Lemma 1.6 of [3], we only need to prove that

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)G(x)\varphi(x-y) dy dx \leq C \|F\|_{L^{2d/(d+\alpha)}(\mathbb{R}^d)} \|G\|_{L^{2d/(d+\alpha)}(\mathbb{R}^d)}, \quad (2.23)$$

where

$$F(x) = \frac{|f(x)|}{(h(x))^{p/2}} \quad \text{and} \quad G(x) = \frac{|g(x)|}{(h(x))^{p/2}},$$

Note that when $\varphi(x) = C|x|^{-(d-\alpha)}$, (2.23) is nothing but (1.27) (*ibid.*). Here we need to handle more general φ as given in (1.11). To prove (2.23), we need to apply the *weak version of Young's inequality* (see, e.g., [10, eq. (7) on p. 107]), which says that for all $p, q, r > 1$ with $1/p + 1/q + 1/r = 2$, it holds that

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x)b(x-y)c(y) dx dy \right| \leq K_{p,q,r,d} \|a\|_{L^p(\mathbb{R}^d)} \|b\|_{q,w} \|c\|_{L^r(\mathbb{R}^d)}, \quad (2.24)$$

where

$$\|b\|_{q,w} := \sup_A |A|^{-1/q'} \int_A |f(x)| dx, \quad \text{with } 1/q + 1/(q') = 1,$$

and A is an arbitrary Borel set of finite measure $|A| < \infty$. Now we apply (2.24) with

$$a = F, \quad c = G, \quad b = \varphi, \quad p = r = 2d/(d+\alpha), \quad \text{and} \quad q = \frac{d}{d - \sum_{j=1}^d \alpha_j} = d/(d-\alpha).$$

By the proof of [1, Lemma 1.6], it suffices to prove that $\|\varphi\|_{q,w}$ is finite with $q = d/(d-\alpha)$ and $1/q' = 1 - 1/q = \alpha/d$.

Recall that according to Assumption 1.2, the d coordinates have been partitioned into k groups. Define $A_R := A_1 \times \cdots \times A_k$ where $A_i = B_{R,d_i}(0)$ is the ball in $\mathbb{R}^{d_i}$ centered at the origin with radius R . With this we have that

$$\int_{A_i} |x_{(i)}|^{-(d_i-\alpha_i)} dx_{(i)} = |\mathbb{S}^{d_i-1}| \frac{R^{\alpha_i}}{\alpha_i}, \quad (2.25) \quad \text{E:int}$$

where we have used polar coordinated to calculate the integral and $|\mathbb{S}^{d_i-1}| = (2\pi^{d_i/2})/(\Gamma(d_i/2))$ is the surface area of the sphere in $\mathbb{R}^{d_i}$ (clearly, when $d_i = 1$, $|\mathbb{S}^0| = 2$). Moreover, by the formula for the volume of balls in $\mathbb{R}^{d_i}$, we see that

$$|A_i| = \frac{\pi^{d_i/2}}{\Gamma(1 + \frac{d_i}{2})} R^{d_i} = |\mathbb{S}^{d_i-1}| \frac{R^{d_i}}{d_i}. \quad (2.26) \quad \text{E:leb}$$

Recall that $1/q' = \alpha/d$. Then a combination of (2.25), (2.26) and (1.11) shows that

$$\begin{aligned} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx &= \prod_{i=1}^k C_{\alpha_i, d_i} |A_i|^{-1/q'} \int_{A_i} |x_{(i)}|^{-(d_i-\alpha_i)} dx_{(i)} \\ &= \prod_{i=1}^k C_{\alpha_i, d_i} \alpha_i^{-1} |\mathbb{S}^{d_i-1}|^{1-\alpha/d} R^{\alpha_i - \frac{d_i}{d} \alpha} d_i^{\alpha/d} \\ &= \prod_{i=1}^k C_{\alpha_i, d_i} \alpha_i^{-1} |\mathbb{S}^{d_i-1}|^{1-\alpha/d} d_i^{\alpha/d} =: K, \end{aligned}$$

where the constants C_{α_i, d_i} are defined in (1.13) and the final constant K does not depend on R . Finally, by symmetry of φ , we have that

$$\|\varphi\|_{q,w} = \sup_{R>0} |A_R|^{-1/q'} \int_{A_R} \varphi(x) dx = K < \infty. \quad (2.27) \quad \text{E:WeakNorm}$$

Hence, $\varphi \in L_{q,w}(\mathbb{R}^d)$ with $q = d/(d - \alpha)$. This completes the proof of Lemma 2.6. $\square$

Remark 2.7. Note that when there is only one partition (i.e., $k = 1$), or equivalently when γ itself is the Riesz kernel, by [10, (6) on p. 107], we see that

$$\left\| |\cdot|^{-(d-\alpha)} \right\|_{\frac{d}{d-\alpha}, w} = \alpha^{-1} |\mathbb{S}^{d-1}|^{1-\alpha/d} d^{\alpha/d},$$

which is consistent with the norm we find in (2.27) up to a constant $C_{\alpha,d}$.

3 Existence and uniqueness of the solution

In this section, we will prove Theorem 1.3. The proof will need the following lemma:

Lemma 3.1 (Lemma 3.5 of [1]). *If $H : [0, \infty) \rightarrow [0, \infty)$ is a non-decreasing function, then*

$$2 \int_0^\infty e^{-2t} H^2(t) dt \leq \left(\int_0^\infty e^{-t} H(t) dt \right)^2. \quad (3.1) \quad \text{E:doubleExp}$$

The proof of Theorem 1.3 follows the same strategy as [1, Section 3] with minor changes such as

$$\frac{1}{1 + |\xi|^2} \quad \text{replaced by} \quad \frac{1}{1 + \frac{\nu}{2} |\xi|^a}. \quad (3.2) \quad \text{E:Changes}$$

Nevertheless, for completeness, here we streamline and reorganize this proof as follows.

Proof of Theorem 1.3. We first introduce some notation. Let $L(x)$ be the Laplace transform of $G(\cdot, x)$ evaluated at one and calculate its Fourier transform $\mathcal{F}L(\xi)$ as follows:

$$L(x) = \int_0^\infty e^{-t} G(t, x) dt \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}; \quad (3.3) \quad \boxed{\text{E:LandFL}}$$

see the proof of Theorem 4.1 of [5] for the last equality. Similarly, let $L_n(\vec{y})$ to be the Laplace transform of $\widetilde{f}_n(\vec{y}, 0, \cdot)$ evaluated at one, namely,

$$\begin{aligned} L_n(\vec{y}) &:= n! \int_0^\infty e^{-t} \widetilde{f}_n(\vec{y}; 0, t) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0, t]^n} \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{s} dt \end{aligned}$$

with the convention that $s_0 = 0$ and $y_{\sigma(0)} = 0$. By the relation of convolution and the Laplace transform (or through a change of variables), we see that

$$L_n(\vec{y}) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)}). \quad (3.4)$$

Hence, from (3.3),

$$\mathcal{F}L_n(\vec{\xi}) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a}. \quad (3.5) \quad \boxed{\text{E:FLfn}}$$

Moreover, define

$$\begin{aligned} H_n(t, \vec{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \widetilde{f}_n(\vec{y}; 0, t) d\vec{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s}. \end{aligned}$$

Under the nonnegativity assumption — Assumption 1.1, we see that for any $\vec{x} \in \mathbb{R}^{nd}$ fixed, the function $t \rightarrow H_n(t, \vec{x})$ is a non-decreasing function for $t \geq 0$. For this function, we are about to apply Lemma 3.1.

Step 1. We first compute the corresponding part to the right-hand side of (3.1). By Fubini's theorem,

$$\begin{aligned} &\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n G(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\vec{y} d\vec{s} dt \\ &= \int_{\mathbb{R}^{dn}} \prod_{k=1}^n K(x_k - y_k) L_n(\vec{y}) d\vec{y}. \end{aligned}$$

Then an application of the Plancherel's theorem and the fact that $K * K = \gamma$ shows that

$$\int_{(\mathbb{R}^d)^n} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x} = \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\vec{\xi})|^2 \mu(d\vec{\xi}); \quad (3.6) \quad \boxed{\text{E:Hneq1}}$$

one may check the proof of Lemma 3.3 of [1] for more details.

Step 2. Now we compute the corresponding part to the left-hand side of (3.1). First, using the fact that $K * K = \gamma$, we see that

$$\left\| \widetilde{f_n}(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \vec{x}) d\vec{x}; \quad (3.7) \quad \text{E:inteq1}$$

one may check the proof of Lemma 3.4 of [1] for more details. By the scaling property for $\mathcal{F}\widetilde{f_n}$ in (2.9), one can show that

$$\int_0^\infty e^{-t} \left\| \widetilde{f_n}(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt; \quad (3.8) \quad \text{E:fHn}$$

see Appendix for the proof.

Step 3. Now we can apply the Fubini theorem and Lemma 3.1 to the function $t \rightarrow H_n(t, \vec{x})$ to see that

$$\int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt \leq \int_{\mathbb{R}^{nd}} \left[\int_0^\infty e^{-t} H_n(t, \vec{x}) dt \right]^2 d\vec{x}. \quad (3.9) \quad \text{E:doubleH}$$

Therefore, combining (3.6), (3.7), and (3.9) shows that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \widetilde{f_n}(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &\leq \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_{(\mathbb{R}^d)^n} |\mathcal{F}L_n(\vec{\xi})|^2 \mu(d\vec{\xi}) \\ &=: \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} T_n(\nu, a), \end{aligned} \quad (3.10) \quad \text{E:fTn}$$

where

$$T_n(\nu, a) = \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a} \right]^2 \mu(d\vec{\xi}). \quad (3.11) \quad \text{E:Tn}$$

By the same arguments as those of Lemma 3.6 of [1] with the replacement (3.2), we see that

$$T_n(\nu, a) \leq (n!)^2 C_\mu^n(\nu, a) \quad \text{with} \quad C_\mu(\nu, a) := \int_{(\mathbb{R}^d)} \left(\frac{1}{1 + \frac{\nu}{2} |\xi|^a} \right)^2 \mu(d\xi) < \infty. \quad (3.12) \quad \text{E:muassumpt}$$

Here, the finiteness of $C_\mu(\nu, a)$ entails one part of the conditions in (1.4):

$$\int_{(\mathbb{R}^d)} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi) = C \int_0^\infty \frac{\rho^{\alpha-1}}{(1 + \rho^a)^2} d\rho < \infty \quad \Longleftrightarrow \quad 2a - \alpha + 1 > 1. \quad (3.13) \quad \text{E:alpha2}$$

Combining (3.10) and (3.12) gives that

$$\int_0^\infty e^{-t} \left\| \widetilde{f_n}(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^n dt \leq 2^{n[2(b+r)-\frac{b\alpha}{a}]} C_\mu^n(\nu, a). \quad (3.14) \quad \text{E:LNfneq}$$

Step 4. From the scaling property (2.10) we see that

$$\int_0^\infty e^{-t} \left\| \widetilde{f_n}(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \left\| \widetilde{f_n}(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t} t^{[2(b+r)-\frac{b\alpha}{a}]} dt$$

$$= \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right), \quad (3.15) \quad \text{E:fnOne}$$

which entails another part of the conditions in (1.4):

$$2(b+r) - \frac{b\alpha}{a} > 0. \quad (3.16) \quad \text{E:alpha3}$$

From (3.14) and (3.15), we deduce that

$$\begin{aligned} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 &= \frac{1}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt \\ &\leq \frac{\left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} C_\mu(\nu, a) \right)^n}{\Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right)} \leq C \frac{\left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} C_\mu(\nu, a) \right)^n}{(n!)^{2(b+r) - b\alpha/a}}, \end{aligned}$$

where the constant C depends only on the value of $2(b+r) - b\alpha/a$. Therefore, by (2.11), (2.10), and the above inequality,

$$\begin{aligned} \|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! t^{[2(b+r) - b\alpha/a]n} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &\leq C \sum_{n \geq 0} \theta^n t^{[2(b+r) - b\alpha/a]n} \frac{\left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} C_\mu(\nu, a) \right)^n}{(n!)^{2(b+r) - 1 - b\alpha/a}}, \end{aligned} \quad (3.17) \quad \text{E:2nomBd}$$

which is finite provided that (see (1.4))

$$2(b+r) - 1 - b\alpha/a > 0. \quad (3.18) \quad \text{E:alpha4}$$

Finally, an application of the Minkowski inequality and the hypercontractivity (see [1, Theorem B.1]) shows that for all $p \geq 2$,

$$\begin{aligned} \|u(t, x)\|_p &\leq \sum_{n \geq 0} \theta^{n/2} (p-1)^{n/2} \sqrt{n!} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}} \\ &\leq \sqrt{C} \sum_{n \geq 0} \theta^{n/2} (p-1)^{n/2} t^{[2(b+r) - b\alpha/a]n/2} \frac{\left(2^{\left[2(b+r) - \frac{b\alpha}{a} \right]} C_\mu(\nu, a) \right)^{n/2}}{(n!)^{\frac{1}{2}[2(b+r) - 1 - b\alpha/a]}}. \end{aligned} \quad (3.19) \quad \text{E:pmomBd}$$

Therefore, under condition (1.4), (1.1) has a unique $L^p(\Omega)$ -solution $u(t, x)$ for all $p \geq 2$, $t > 0$ and $x \in \mathbb{R}^d$. This proves part (1) of Theorem 1.3.

Step 5. The proof of part (2) of Theorem 1.3 will be postponed to part (ii) of Lemma 4.1 below. $\square$

4 Upper bound of the asymptotics

In this section, we will establish the upper bound of (1.7) (under Assumptions 1.1 and 1.2, and condition (1.4)), namely,

$$\begin{aligned} \limsup_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\leq \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ &\quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right). \end{aligned} \quad (4.1) \quad \text{E:momentasy}$$

We will first establish the corresponding result for $p = 2$ in Lemma 4.2 and then apply the hypercontractivity property given by Theorem B.1 in [1] to obtain the general case for $p \geq 2$. To prove the next lemma, we will need the following equality

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma(an + 1)}{(n!)^a} \right) = a \log(a), \quad \text{for all } a > 0, \quad (4.2) \quad \text{E:Stirling}$$

which is a direct consequence of Stirling's formula.

L:2ineq

Lemma 4.1. *Assume Assumptions 1.1 and 1.2 hold.*

(i) *Under condition (1.4),*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right) = \log \left(2^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad \text{and} \quad (4.3) \quad \text{E:2ineq-a}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2(b+r) - \frac{b\alpha}{a}} \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \right) + \log \rho. \quad (4.4) \quad \text{E:2ineq-b}$$

(ii) *Under condition (1.5), for all $x \in \mathbb{R}^d$, $p \geq 2$ and $\theta > 0$, it holds that*

$$\|u(t, x)\|_p < \infty \quad \text{whenever } t < T_p, \quad (4.5) \quad \text{E:pnormconv}$$

where T_p be defined as in (1.6). Moreover,

$$\sum_{n \geq 0} \theta^n n! \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \begin{cases} < \infty & \text{if } t < T_2, \\ = \infty & \text{if } t > T_2. \end{cases} \quad (4.6) \quad \text{E:2normcut}$$

Proof. (i) The proof of part (i) follows the same arguments as those of [1, Lemma 4.3]. Nevertheless, we sketch the proof here for completeness. Recall the definition of $T_n(\nu, a)$ defined in (3.11). From (2.22), we see that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{T_n(\nu, a)}{(n!)^2} \right] = \log(\rho_{\nu, a}). \quad (4.7) \quad \text{E:LimTn}$$

As a consequence of (3.10) and (3.15) in the proof of Theorem 1.3, we see that

$$\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n(\nu, a)}{(n!)^2}.$$

Combining this and (2.22) we see that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \log & \left[\left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \right] \\ & \leq \log \left(2^{[2(b+r) - \frac{b\alpha}{a}]} \right) + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{T_n(\nu, a)}{(n!)^2} \right) \\ & = \log \left(2^{[2(b+r) - \frac{b\alpha}{a}]} \right) + \log(\rho_{\nu, a}), \end{aligned}$$

which proves the upper bound for (4.3).

Now we prove the lower bound for (4.3). Let τ and $\tilde{\tau}$ be independent exponential random variables with mean one. In the following, we will compute $\mathbb{E}[J_n(\tau, \tilde{\tau})]$ in two ways, where

$$J_n(t, t') := \int_{\mathbb{R}^{nd}} H_n(t, x) H_n(t', x) dx, \quad t, t' > 0.$$

Notice that using the above notation, (3.7) can be rewritten as

$$\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n(t, \vec{x})^2 d\vec{x} = \frac{1}{(n!)^2} J_n(t, t).$$

On the one hand, the Cauchy-Schwartz inequality implies that

$$J_n(t, t') \leq J_n(t, t)^{1/2} J_n(t', t')^{1/2} = t^{[2(b+r)-b\alpha/a]n/2} (t')^{[2(b+r)-b\alpha/a]n/2} J_n(1, 1).$$

where we have used the scaling property of $J_n(t, t)$ inherited from that of $\left\| \widetilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}$ as in (2.10). Hence,

$$\begin{aligned} \mathbb{E}[J_n(\tau, \tilde{\tau})] &\leq \mathbb{E} \left[\tau^{[2(b+r)-b\alpha/a]n/2} \right] \mathbb{E} \left[\tilde{\tau}^{[2(b+r)-b\alpha/a]n/2} \right] J_n(1, 1) \\ &= \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 (n!)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \end{aligned}$$

On the other hand, by (3.6) and (3.11), we see that

$$\mathbb{E}[J_n(\tau, \tilde{\tau})] = \int_0^\infty \int_0^\infty e^{-t} e^{-\tilde{t}} J_n(t, \tilde{t}) dt d\tilde{t} = \int_{\mathbb{R}^{dn}} \left[\int_0^\infty e^{-t} H_n(t, x) dt \right]^2 dx = T_n(\nu, a).$$

Therefore,

$$\frac{T_n(\nu, a)}{(n!)^2} \leq \Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \left\| \widetilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2. \quad (4.8) \quad \boxed{\text{E:2lowbd}}$$

Now, an application of Stirling's formula shows that, as $n \rightarrow \infty$,

$$\Gamma \left(\frac{[2(b+r)-b\alpha/a]n}{2} \right)^2 \sim \Gamma([2(b+r)-b\alpha/a]n+1) 2^{-[2(b+r)-b\alpha/a]n} C_n, \quad (4.9) \quad \boxed{\text{E:GammaSq}}$$

where $C_n = 2^{-1}[2(b+r)-b\alpha/a]^{1/2}(2\pi n)^{1/2}$. Then an application of (4.7), (4.8) and (4.9) proves the lower bound of (4.3). Lastly, (4.4) follows from (4.3) and the limit (4.2). This proves part (i) of Lemma 4.1.

(ii) Note that part (ii) is equivalent to the statement of part (2) of Theorem 1.3. The critical case happens when the exponent of $n!$ in (3.19) vanishes, namely,

$$\alpha = \frac{a}{b} [2(b+r) - 1].$$

Among the three inequalities in (1.4), we also need to make sure that the minimum is achieved by $\frac{a}{b} [2(b+r) - 1]$, for which, we need to additionally require $\frac{a}{b} [2(b+r) - 1] \leq d$ and

$$\frac{a}{b} [2(b+r) - 1] \leq 2a \iff r \in [0, 1/2].$$

Putting these conditions together gives the conditions stated in (1.5).

We start by proving (4.5). Let u_λ for $\lambda > 0$ be the solution of the SPDE (1.1) with θ replaced with λ and $u = u_\theta$. By the hyperactivity property (see [1, Lemma B.1]), we have that

$$\|u(t, x)\|_p \leq \|u_{(p-1)\theta}(t, x)\|_2, \quad \text{for all } p \geq 2, t > 0 \text{ and } x \in \mathbb{R}^d. \quad (4.10) \quad \boxed{\text{E:HyContr}}$$

Now by recalling Theorem 2.3 and by applying $\alpha = \frac{a}{b} [2(b+r) - 1]$ and the scaling property (2.10), we see that

$$\begin{aligned} \|u_{(p-1)\theta}(t, x)\|_2^2 &= \sum_{n \geq 0} [\theta(p-1)]^n n! \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} [t\theta(p-1)]^n n! \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &=: \sum_{n \geq 0} [t\theta(p-1)]^n R_n, \end{aligned}$$

with $R_n = n! \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2$. By the Cauchy-Hadamard theorem, this series converges for $\theta t(p-1) < \limsup_{n \rightarrow \infty} |R_n|^{-1/n}$. However, by (4.4) and Lemma 2.4, we see that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log(|R_n|) = \log(2\rho) = \log\left(2\nu^{-\alpha/a} \mathcal{M}_a^{(2a-\alpha)/a}\right).$$

Therefore, $\limsup_{n \rightarrow \infty} |R_n|^{-1/n} = (2\nu^{-\alpha/a} \mathcal{M}_a^{(2a-\alpha)/a})^{-1}$ and $\|u(t, x)\|_p$ converges for

$$t < \frac{1}{2\theta(p-1)\nu^{-\alpha/a} \mathcal{M}_a^{(2a-\alpha)/a}} =: T_p; \quad \text{see (1.6).}$$

To show (4.6), we use the Cauchy-Hadamard theorem and the same techniques above to see that the radius of convergence of the series

$$\|u(t, x)\|_2^2 = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2,$$

is precisely T_2 . This completes the proof of Lemma 4.1. $\square$

L:p=2 **Lemma 4.2.** Assume Assumptions 1.1 and 1.2 hold. Under condition (1.4), we have that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) = \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta (\theta\rho)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right).$$

Proof. By part (1) of Theorem 1.3, there is an $L^2(\Omega)$ solution $u(t, x)$. By the scaling property (2.10),

$$\begin{aligned} \mathbb{E}(|u(t, x)|^2) &= \sum_{n \geq 0} \theta^n (n!) \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n (n!) t^{(2(b+r) - b\alpha/a)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} z_n R_n t^{(2(b+r) - b\alpha/a)n} \end{aligned}$$

where

$$R_n = (n!)^{2(b+r) - b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad z_n = \frac{\theta^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}}.$$

Notice that (4.4) above says that

$$\frac{1}{n} \log(R_n) \rightarrow \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad \text{as } n \rightarrow \infty.$$

Now define R to be

$$R = \left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho.$$

We want to find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} z_n R_n \left(t^{2(b+r) - b\alpha/a} \right)^n = A.$$

Indeed, by the following limit (see [1, Lemma A.3])

$$\lim_{t \rightarrow \infty} t^{-1/\gamma} \log \sum_{n \geq 0} (n!)^{-\gamma} t^n = \gamma, \quad \text{for all } \gamma > 0,$$

we see that

$$\lim_{t \rightarrow \infty} \left[\frac{1}{(\theta R) t^{2(b+r) - b\alpha/a}} \right]^{\frac{1}{2(b+r) - (b\alpha/a) - 1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r) - b\alpha/a}]^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}} = 2(b+r) - (b\alpha/a) - 1,$$

which, by an easy algebraic manipulation, is equivalent to

$$\begin{aligned} & \lim_{t \rightarrow \infty} \left[\frac{1}{t^{2(b+r) - b\alpha/a}} \right]^{\frac{1}{2(b+r) - (b\alpha/a) - 1}} \log \sum_{n \geq 0} \frac{[(\theta R) t^{2(b+r) - b\alpha/a}]^n}{(n!)^{2(b+r) - (b\alpha/a) - 1}} \\ &= [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{1}{2(b+r) - (b\alpha/a) - 1}}. \end{aligned}$$

Hence,

$$\beta = \frac{2a(b+r) - b\alpha}{2a(b+r) - (b\alpha) - a} \quad \text{and} \quad A = [2(b+r) - (b\alpha/a) - 1] (\theta R)^{\frac{a}{2a(b+r) - b\alpha - a}}.$$

Finally, an application of [1, Lemma A.2] shows that

$$\begin{aligned} & \lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \mathbb{E}(|u(t, x)|^2) \\ &= \theta^{\frac{a}{2a(b+r) - b\alpha - a}} \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \rho^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \end{aligned}$$

which proves Lemma 4.2. □

Now we are ready to prove (4.1).

Proof of (4.1). By the hypercontractivity (4.10) and the scaling property (2.10), we see that for all $p \geq 2$,

$$\begin{aligned} \|u(t, 0)\|_p^2 &\leq \|u_{(p-1)\theta}(t, 0)\|_2^2 = \sum_{n \geq 0} (n!)^n \theta^n (p-1)^n \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (n!)^n \theta^n \left\| \tilde{f}_n \left(\cdot, 0, t(p-1)^{\frac{1}{2(b+r) - b\alpha/a}} \right) \right\|_{\mathcal{H}^{\otimes n}}^2 \\ &= \left\| u \left(t(p-1)^{\frac{1}{2(b+r) - b\alpha/a}}, 0 \right) \right\|_2^2. \end{aligned}$$

Hence,

$$\|u(t, 0)\|_p \leq \left\| u \left(t(p-1)^{\frac{1}{2(b+r) - b\alpha/a}}, 0 \right) \right\|_2 =: \|u(t_p, 0)\|_2, \quad (4.11) \quad \boxed{\text{E:pMom2}}$$

where t_p is defined in (1.8). Finally, an application of Lemma 4.2 proves (4.1). □

5 Lower bound of the asymptotics

S:lowerbd

In this section, we will prove the lower bound of (1.7), namely,

$$\liminf_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \geq \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \quad (5.1)$$

E:momentasy

Through out this section, we assume that Assumptions 1.1 and 1.2, and condition (1.4) hold.

We start by defining the function $W_n(t, \phi)$ on $(0, \infty) \times L_{\mathbb{C}}^2(\mu)$ by

$$W_n(t, \phi) := \int_{[0, t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1}, \cdot)(\xi_k + \dots + \xi_n) \mu(d\xi_1) \dots \mu(d\xi_n) ds. \quad (5.2)$$

with $s_0 = 0$. We now give conditions under which W_n is well defined.

L:LWn

Lemma 5.1. *If the measure μ satisfies the relation in (3.12), then $W_n(t, \phi)$ is well defined and for any $d \geq 1$, $t > 0$ and $\phi \in L_{\mathbb{C}}^2(\mu)$. Moreover,*

$$\int_0^\infty e^{-t} W_n(t, \phi) dt = \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi) \dots \mu(d\xi_n). \quad (5.3)$$

E:LWn

Proof. The proof follows the exact arguments as those in the proof of Lemma 6.2 of [1] except that one needs to use the following Laplace transform:

$$\int_0^\infty e^{-t} \mathcal{F}G(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a};$$

see the second equation in (3.3). □

The next Proposition is a restatement of Proposition 6.3 of [1]. The proof follows the same proof as that of Proposition 6.3 of [1]. We will not repeat here.

P:Wnprop1

Proposition 5.2. *For $f \in \mathcal{H}$, $t > 0$, and $p, q > 1$ with $p^{-1} + q^{-1} = 1$, it holds that*

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} (q-1) \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right| \quad (5.4)$$

E:Wnprop1

and as a consequence, the series $\left| \sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f) \right|$ converges provided that $\|u(t, 0)\|_p < \infty$, which is the case under Theorem 1.3.

Now we are going to apply a scaling argument to (5.4) in order to put t and p together, from which we can determine the constants t_p and β defined in (1.8).

P:Wnprop2

Proposition 5.3. *For $p, q > 1$ and $p^{-1} + q^{-1} = 1$ we have that*

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n \left(t_p^\beta, \mathcal{F}f \right) \right|, \quad (5.5)$$

E:Wnprop2

where the constants β and t_p are defined in (1.8).

Proof. From Proposition 5.2, we see that for any $f \in \mathcal{H}$, the inequality (5.4) holds. For some constants $V, W > 0$, which will be determined in this proof, let $f_\tau(x) := \tau^V f(\tau^W x)$ be a scaled version of f . It is clear that $f_\tau \in \mathcal{H}$. By some elementary scaling arguments (see the proof of the Lemma 6.4 of [1]), one can show that

$$\|f_\tau\|_{\mathcal{H}}^2 = \tau^{2(V-dW)+W\alpha} \|f\|_{\mathcal{H}}^2 \quad \text{and} \quad (5.6)$$

$$W_n(t, \mathcal{F}f_\tau) = \tau^n [V-W((d-\alpha)+\frac{a}{b}(b+r))] W_n\left(t\tau^{\frac{a}{b}W}, \mathcal{F}f\right). \quad (5.7)$$

Hence, an application of Proposition 5.2 to f_τ shows that

$$\begin{aligned} \|u(t, 0)\|_p &\geq \exp\left\{-\frac{1}{2}(q-1)\|f_\tau\|_{\mathcal{H}}^2\right\} \left|\sum_{n \geq 0} \theta^{n/2} W_n(t, \mathcal{F}f_\tau)\right| \\ &= \exp\left\{-\frac{1}{2}(q-1)\tau^{2(V-dW)+W\alpha}\|f\|_{\mathcal{H}}^2\right\} \left|\sum_{n \geq 0} \theta^{n/2} \tau^n [V-W((d-\alpha)+\frac{a}{b}(b+r))] W_n\left(t\tau^{\frac{a}{b}W}, \mathcal{F}f\right)\right|. \end{aligned}$$

Comparing the above lower bound with that in (5.5), we obtain the following two relations with three unknowns W , V and τ :

$$V - W\left((d-\alpha) + \frac{a}{b}(b+r)\right) = 0, \quad (5.8) \quad \boxed{\text{E:VW-1}}$$

$$(p-1)^{-1} \tau^{2(V-dW)+W\alpha} = t\tau^{\frac{a}{b}W}. \quad (5.9) \quad \boxed{\text{E:VW-2}}$$

Since (5.9) should hold for all $t > 0$ and $p \geq 2$, one can choose $\tau = (p-1)t$ to reduce the relation (5.9) to

$$\tau^{2(V-dW)+W\alpha} = \tau^{\frac{a}{b}W+1},$$

which then gives the following equation

$$2(V-dW) + W\alpha = 1 + \frac{a}{b}W. \quad (5.10) \quad \boxed{\text{E:VW-3}}$$

Now solve the linear equations (5.8) and (5.10) for W and V to see that

$$\begin{cases} W = \frac{b}{a}(\beta-1), \\ V = \left(\frac{a}{b}(b+r) - \alpha + d\right) \frac{b}{a}(\beta-1), \end{cases} \quad \text{with } \beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}.$$

Therefore, the scaling f_τ and t_p^β should be

$$f_\tau(x) = \tau^{\left(\frac{a}{b}(b+r) - \alpha + d\right) \frac{b}{a}(\beta-1)} f\left(\tau^{\frac{b}{a}(\beta-1)} x\right) \quad \text{and} \quad t_p^\beta := t\tau^{\frac{a}{b}W} = (p-1)^{\beta-1} t^\beta,$$

respectively. This completes the proof of Proposition 5.3. $\square$

To remove the absolute value sign in Proposition 5.3, we want to identify all the $f \in \mathcal{H}$ for which $W_n(t, \mathcal{F}f)$ is nonnegative. In fact, if we consider the space

$$\mathcal{H}_+ = \{f \in \mathcal{H} : f \text{ is nonnegative and nonnegative definite}\},$$

then by Plancherel's theorem, for $f \in \mathcal{H}_+$,

$$W_n(t, \mathcal{F}f) = \int_{[0,t]^n} \int_{\mathbb{R}^{nd}} \prod_{k=1}^n (f * \gamma)(x_k) \prod_{k=1}^n G(s_k - s_{k-1}, x_k - x_{k-1}) d\vec{s} =: U_n(t, f) \geq 0$$

with the convention that $s_0 = 0$ and $x_0 = 0$, where we have used the fact that the fundamental solution $G(t, x)$ is nonnegative (under Assumption 1.1). Now define

$$W_n(t) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} W_n(t, \mathcal{F}(f)) \quad \text{and} \quad U_n(t) := \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} U_n(t, f).$$

It is clear that $W_n(t) \geq U_n(t) \geq 0$.

P:LogU

Proposition 5.4. *If τ is an exponential random variable with mean one, then*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log(\rho^{1/2}).$$

Proof. We start by letting τ be an exponential random variable with mean one. With this,

$$\mathbb{E}(U_n(\tau)) = \int_0^\infty e^{-t} U_n(t) dt \geq \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \int_0^\infty e^{-t} U_n(t, f) dt.$$

For any $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = 1$, by Bochner's theorem, $\mathcal{F}f$ is nonnegative and nonnegative definite, which further implies that $\mathcal{F}f$ is even. In addition, Lemma 5.1 gives us that

$$\begin{aligned} \int_0^\infty e^{-t} U_n(t, f) dt &= \int_0^\infty e^{-t} W_n(t, \mathcal{F}f) dt \\ &= \int_{\mathbb{R}^{dn}} \prod_{k=1}^n \mathcal{F}f(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \cdots \mu(d\xi_n). \end{aligned}$$

By the same arguments as [1, Assumption C] with the replacement (3.2) (see also the proof of [3, Theorem 1.2]), we have that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \int_{\mathbb{R}^{dn}} \prod_{k=1}^n (\mathcal{F}f)(\xi_k) \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\xi_k + \dots + \xi_n|^a} \mu(d\xi_1) \cdots \mu(d\xi_n) \geq \log \rho(\mathcal{F}f),$$

where $\rho(\cdot)$ is defined on the space of nonnegative and nonnegative definite functions and is given by

$$\rho(g) := \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} g(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta) h(\eta)}{\sqrt{1 + \frac{\nu}{2} |\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2} |\eta|^a}} d\eta \right] \mu(d\xi).$$

Note that since both μ and g are nonnegative, we may assume h is also nonnegative for the remainder of this proof. With this being said, we see that for any $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = 1$,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log \rho(\mathcal{F}f).$$

We now need to calculate

$$\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f).$$

Consider a nonnegative function $h \in L^2(\mathbb{R}^d)$. The function $h(\cdot)/\sqrt{1 + \frac{\nu}{2}|\cdot|^a} \in L^2(\mathbb{R}^d)$ so that $g_h(x) := (2\pi)^{d/2} \mathcal{F}^{-1} \left(\frac{h(\cdot)}{\sqrt{1 + \frac{\nu}{2}|\cdot|^a}} \right) (x)$ is well defined. Under these conditions, $g \in W^{1,a}(\mathbb{R}^d)$ with

$$\|g\|_{W^{1,a}(\mathbb{R}^d)} = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (1 + |\xi|^a) |\mathcal{F}g(\xi)|^2 d\xi = \int_{\mathbb{R}^d} \frac{1 + |\xi|^a}{1 + \frac{\nu}{2}|\xi|^a} |h(\xi)|^2 d\xi \leq C_\nu \|h\|_{L^2(\mathbb{R}^d)}^2 < \infty.$$

Notice that

$$\begin{aligned} \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}g_h(\xi - \eta) \mathcal{F}g_h(-\eta) d\eta \\ &= (2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi), \end{aligned}$$

where we have used the notation that $\widetilde{h}(x) = h(-x)$. Since $h(\cdot)$ is real valued, we see that $\widetilde{\mathcal{F}g_h} = \overline{\mathcal{F}g_h} = \mathcal{F}\bar{g}_h$. Using this and the fact that $\mathcal{F}(fg) = (2\pi)^{-d} \mathcal{F}(f) * \mathcal{F}(g)$, we see that $(2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) = \mathcal{F}[|g_h|^2](\xi)$. Hence,

$$\begin{aligned} \rho(\mathcal{F}f) &= \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \mathcal{F}[|g_h|^2](\xi) \mu(d\xi) \\ &= \sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}}. \end{aligned}$$

With this we see that

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \sup_{\|h\|_{L^2}=1} \langle f, |g_h|^2 \rangle_{\mathcal{H}} = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2} \quad (\text{E:supf}) \quad (5.11)$$

where the optimal f is chosen to be $|g_h|^2 / \| |g_h|^2 \|_{\mathcal{H}}$. Now we claim that

$$|g_h|^2 \in \mathcal{H}_+, \quad (\text{E:GH+}) \quad (5.12)$$

which implies that the supremum in (5.11) can be restricted to $\mathcal{H}_+$ and

$$\sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \rho(\mathcal{F}f) = \sup_{\|h\|_{L^2}=1} \langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}}^{1/2}.$$

Then notice that

$$\mathcal{F}(|g_h|^2)(\xi) = (2\pi)^{-d} (\mathcal{F}g_h * \widetilde{\mathcal{F}g_h})(\xi) = \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + \frac{\nu}{2}|\xi + \eta|^a} \sqrt{1 + \frac{\nu}{2}|\eta|^a}} d\eta. \quad (\text{E:gsq}) \quad (5.13)$$

Hence, from (2.12) and (5.13), we see that

$$\langle |g_h|^2, |g_h|^2 \rangle_{\mathcal{H}} = \int_{\mathbb{R}^d} |\mathcal{F}(|g_h|^2)(\xi)|^2 \mu(d\xi) = \rho_{\nu,a}$$

which then leads us to the lower bound:

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}(U_n(\tau)) \geq \log \left(\rho_{\nu,a}^{1/2} \right).$$

Therefore, it remains to proving (5.12). First notice that from the above we see that

$$\int_{\mathbb{R}^d} |\mathcal{F}(|g_h|^2)(\xi)|^2 \mu(d\xi) \leq \rho_{\nu,a} < \infty \implies |g_h(\cdot)|^2 \in \mathcal{H}.$$

Moreover, since h is nonnegative, from (5.13), we see that $\mathcal{F}(|g_h|^2)(\cdot)$ is also nonnegative. The Bochner-Schwarz theorem then implies that $|g_h(\cdot)|^2$ is nonnegative definite. It is clear that $|g_h(\cdot)|^2$ is nonnegative. This shows that $|g_h(\cdot)|^2 \in \mathcal{H}_+$. This completes the whole proof of Proposition 5.4. $\square$

L:LogU

Lemma 5.5. *It holds that*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[(n!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1) \right] \geq \log \left(\frac{\rho_{\nu,a}^{1/2}}{\left(\frac{b}{a} \left[a - \frac{\alpha}{2} + \frac{a}{b}r \right] \right)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right). \quad (5.14) \quad \text{E:LogU}$$

Proof. Let τ be an exponential random variable with mean one. Notice that by some elementary scaling arguments, we have that for all $t > 0$,

$$W_n(t) = t^{n\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} W_n(1) \quad \text{and} \quad U_n(t) = t^{n\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1), \quad (5.15) \quad \text{E:WU}$$

which then imply that

$$\mathbb{E}[U_n(\tau)] = \mathbb{E} \left(\tau^{n\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} \right) U_n(1) = \Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1).$$

Then by Proposition 5.4,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left[\Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right) U_n(1) \right] \geq \log \left(\rho_{\nu,a}^{1/2} \right). \quad (5.16) \quad \text{E:1b1}$$

Therefore, (5.14) is proved by noticing that, thanks to (4.2),

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{\Gamma \left(n\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) + 1 \right)}{(n!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right) = \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log \left(\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right),$$

where the condition $\frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) > 0$ is guaranteed by (3.18) (or (1.4)). $\square$

We need one last lemma before the proof of the lower bound:

Lemma 5.6. *For any $k, \theta > 0$, there exists a constant $c_1 = c_1(\alpha, \mathcal{M}, k, \theta) > 0$ such that, by setting $n_t = \lfloor c_1 t \rfloor$, it holds that*

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log(k^{n_t} \theta^{n_t/2} U_{n_t}(t)) \geq \left(k\sqrt{\theta} \right)^{\frac{2a}{2ab - \alpha b + 2ar}} \left(\rho_{\nu,a}^{1/2} \right)^{\frac{2a}{2ab - \alpha b + 2ar}}. \quad (5.17) \quad \text{E:69ineq}$$

Proof. Fix an arbitrary $\epsilon > 0$. Lemma 5.5 guarantees the existence of an $N_\epsilon > 0$ so for all $n > N_\epsilon$

$$(n!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_n(1) \geq \exp(n(\log(R) - \epsilon)) = R^n e^{-n\epsilon} \quad (5.18) \quad \text{E:LogUnlb-1}$$

where

$$R = \rho_{\nu,a}^{1/2} \left(\frac{b}{a} \left[a - \frac{\alpha}{2} + \frac{a}{b}r \right] \right)^{-\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}.$$

Now fix a $c > 0$ and let $n_t := \lfloor ct \rfloor$. Notice that $n_t \geq N_\epsilon$ for any $t > t_\epsilon := (N_\epsilon + 1)/c$. For $t > t_\epsilon$, from (5.18), we have

$$k^{n_t} \theta^{n_t/2} U_{n_t}(t) = k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} U_{n_t}(1) \geq \frac{k^{n_t} \theta^{n_t/2} t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} R^{n_t} e^{-n_t \epsilon}}{(n_t!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}}. \quad (5.19) \quad \boxed{\text{E:69ineq2}}$$

Notice that $\lfloor ct \rfloor / t \rightarrow c$ as $t \rightarrow \infty$ which means that $n_t / t \rightarrow c$ as $t \rightarrow \infty$. With this we can say

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) = c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(k^{n_t} \theta^{n_t/2} U_{n_t}(t) \right) =: I(n_t).$$

Now, by (5.19), we have that

$$\begin{aligned} I(n_t) &\geq c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left((kR\sqrt{\theta})^{n_t} \frac{t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} n_t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}}{(n_t!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) + c \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left[\left(\frac{t}{n_t} \right)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)} \frac{n_t^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}}{(n_t!)^{\frac{b}{a}(a - \frac{\alpha}{2} + \frac{a}{b}r)}} \right] - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \liminf_{t \rightarrow \infty} \frac{1}{n_t} \log \left(\frac{n_t^{n_t}}{(n_t)!} \right) - c\epsilon \\ &= c \log(k\sqrt{\theta}R) - c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + c \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) - c\epsilon \end{aligned}$$

and letting ϵ tend to 0 we see that

$$I(n_t) \geq c \left[\log(k\sqrt{\theta}R) - \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \log(c) + \frac{b}{a} \left(a - \frac{\alpha}{2} + \frac{a}{b}r \right) \right] =: h(c).$$

In order to maximize $h(c)$, notice that

$$h'(c) = 0 \quad \Longleftrightarrow \quad c^* = (k\sqrt{\theta}R)^{\frac{a}{b(a - \frac{\alpha}{2} + \frac{a}{b}r)}}.$$

After plugging c^* and replacing R we arrive at the following inequality

$$I(n_t) \geq (k\sqrt{\theta})^{\frac{a}{b(a - \frac{\alpha}{2} + \frac{a}{b}r)}} (\rho_{\nu,a}^{1/2})^{\frac{a}{b(a - \frac{\alpha}{2} + \frac{a}{b}r)}},$$

which proves (5.17) after some simplification. $\square$

We are now ready to prove (5.1).

Proof of (5.1). By proposition 5.3, for and $p, q > 0$ with $p^{-1} + q^{-1} = 1$ we have that

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta \|f\|_{\mathcal{H}}^2 \right\} \left| \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \right|.$$

We now take the supremum over all $f \in \mathcal{H}_+$ with $\|f\|_{\mathcal{H}} = k > 0$. Recall that $W_n(t, \phi) = k^n W_n(t, \phi/k)$ for $\phi \in L^2(\mu)$ and non-negativity of $W_n(t, \cdot)$ on $\mathcal{H}_+$. Let $c > 0$.

$$\|u(t, 0)\|_p \geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}} = k} \sum_{n \geq 0} \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f)$$

$$\begin{aligned}
&= \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} \sum_{n \geq 0} k^n \theta^{n/2} W_n(t_p^\beta, \mathcal{F}f) \\
&\geq \exp \left\{ -\frac{1}{2} t_p^\beta k^2 \right\} \sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f)
\end{aligned}$$

where $n_t = \lfloor ct_p^\beta \rfloor$. Now by choosing c as in Lemma 5.6 we get that

$$\begin{aligned}
\liminf_{n \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p &\geq \liminf_{n \rightarrow \infty} \left(\frac{-\frac{1}{2} t_p^\beta k^2}{t_p^\beta} + \frac{1}{t_p^\beta} \log \left[\sup_{f \in \mathcal{H}_+, \|f\|_{\mathcal{H}}=1} k^{n_t} \theta^{n_t/2} U_{n_t}(t_p^\beta, f) \right] \right) \\
&= -\frac{1}{2} k^2 + k^{\frac{2a}{2ab-\alpha b+2ar}} (\rho \theta)^{\frac{a}{2ab-\alpha b+2ar}} =: h(k).
\end{aligned}$$

By maximizing h for $k > 0$, we see that h is maximized at the point

$$k^* = \left(\frac{2ab - \alpha b + 2ar}{2aB} \right)^{\frac{2ab-\alpha b+2ar}{2a-2[2ab-\alpha b+2ar]}} \quad \text{with } B = (\theta \rho)^{\frac{a}{2ab-\alpha b+2ar}}.$$

Inserting k^* into h gives us that

$$h(k^*) = B^\beta \left(\frac{2a}{2ab - \alpha b + 2ar} \right)^\beta \left(\frac{2ab - \alpha b + 2ar - a}{2a} \right)$$

and plugging in the value for B proves (5.1). $\square$

6 Appendix

Proof of Lemma 2.4. Recall that we define $\varphi(x) = C_{\alpha,d} |x|^{-(d-\alpha)}$. We begin by applying the substitution $x' = (\frac{\nu}{2})^{1/a}$ to ρ :

$$\begin{aligned}
\rho &= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx) \\
&= \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \left(\frac{\nu}{2}\right)^{(d-\alpha)/a} \varphi(x) \left(\frac{\nu}{2}\right)^{-d/a} dx \\
&= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f\left(\left(\frac{\nu}{2}\right)^{-1/a}(x+y)\right) f\left(\left(\frac{\nu}{2}\right)^{-1/a}y\right)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} \left(\frac{\nu}{2}\right)^{-d/a} dy \right]^2 \varphi(x) dx
\end{aligned}$$

Now define $f^*(x) = \left(\frac{\nu}{2}\right)^{\frac{-d}{2a}} f\left(\left(\frac{\nu}{2}\right)^{-1/a}x\right)$. Then with this we see that

$$\rho = \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx$$

and note that

$$\int_{\mathbb{R}^d} f(x)^2 dx = \int_{\mathbb{R}^d} f^*(x)^2 dx$$

and thus

$$\left\{f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1\right\} = \left\{f \in L^2(\mathbb{R}^d) : \|f^*\|_2 = 1\right\}$$

so we can say that

$$\begin{aligned} \rho &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \sup_{\|f^*\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f^*(x+y) f^*(y)}{\sqrt{1+|x+y|^a} \sqrt{1+|y|^a}} dy \right]^2 \varphi(x) dx \\ &= \left(\frac{\nu}{2}\right)^{-\alpha/a} \rho_{2,a} \end{aligned}$$

and (2.19) is proven by applying (2.18).

To prove (2.22), we first notice that by applying the substitution $\xi'_{\sigma(j)} = (\frac{\nu}{2})^{1/a} \xi_{\sigma(j)}$ that

$$\begin{aligned} &\int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \\ &= \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \end{aligned}$$

and now we can directly calculate

$$\begin{aligned} &\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \left(\frac{\nu}{2}\right)^{-n\alpha/a} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \right] \\ &= \log \left[\left(\frac{\nu}{2}\right)^{-\alpha/a} \right] + \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right]^2 \mu(d\vec{\xi}) \right] \end{aligned}$$

and because of both Theorem 1.5 [3] and (2.19) above,

$$\begin{aligned} &\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right) \\ &= \log \left(\left(\frac{\nu}{2}\right)^{-\alpha/a} \right) + \log(\rho_{2,a}) = \log(\rho_{\nu,a}). \end{aligned}$$

This completes the proof of Lemma 2.4. □

Proof of (3.8). Because

$$\mathcal{F}\tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \mathcal{F}f_n(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)}),$$

by the linearity, it suffices to just look at the $\mathcal{F}f(\cdot, 0, ct)(\xi_{\sigma(1)}, \dots, \xi_{\sigma(n)})$ terms individually. For any $c > 0$, by (2.9), we have that

$$\begin{aligned}\mathcal{F}f_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) &= \int_{[0, ct]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(t_{k+1} - t_k, \cdot) \left(\sum_{j=1}^k \xi_j \right)} d\vec{t} \\ &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot) \left(\sum_{j=1}^k \xi_j \right)} c^n d\vec{t}\end{aligned}$$

where we applied the substitution $t'_i = t_i/c$. By the scaling property of $\mathcal{F}G$ we see that

$$\begin{aligned}\int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{\mathcal{F}G(c(t_{k+1} - t_k), \cdot) \left(\sum_{j=1}^k \xi_j \right)} c^n d\vec{t} &= \int_{[0, t]_{<}^n} \prod_{k=1}^n \overline{c^{b+r-1} \mathcal{F}G(t_{k+1} - t_k, \cdot) \left(c^{b/a} \sum_{j=1}^k \xi_j \right)} c^n d\vec{t} \\ &= c^{n(b+r)} \mathcal{F}f_n(\cdot, 0, t)(c^{b/a} \xi_1, \dots, c^{b/a} \xi_n).\end{aligned}$$

Now by the above scaling property, we can obtain (3.8) as follows:

$$\begin{aligned}\int_0^\infty e^{-t} \left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \int_0^\infty e^{-t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \int_0^\infty e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0; 2t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) 2 dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0, t)(2^{b/a} \xi_1, \dots, 2^{b/a} \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= 2^{2n(b+r)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 2^{-\frac{nb d}{a}} 2^{\frac{nb(d-\alpha)}{a}} \mu(d\vec{\xi}) dt \\ &= 2^{n(2(b+r)-b\alpha/a)} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} \left| \mathcal{F} \widetilde{f}_n(\cdot, 0; t)(\xi_1, \dots, \xi_n) \right|^2 \mu(d\vec{\xi}) dt \\ &= \frac{2^{n(2(b+r)-b\alpha/a)}}{(n!)^2} \int_0^\infty 2e^{-2t} \int_{\mathbb{R}^{nd}} H_n(t, \vec{x})^2 d\vec{x} dt.\end{aligned}$$

proves (3.8). □

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